

Databases, Storage Systems & Productivity Tools

Concepts and Applications

DATABASES

Organize information

STORAGE

Keep information

PRODUCTIVITY

Use information

Learning Objectives

By the end of this lesson, you should be able to explain how databases, storage systems, and productivity tools support common digital tasks at school, at work, and in everyday life.

By the end, you should be able to...	In simple words
Explain what a database does	Tell why a school, bank, shop, or app needs organized data.
Identify tables, records, and fields	Look at a table and point out a row, a column, and one value.
Compare SQL and NoSQL	Explain when fixed tables are useful and when flexible data is useful.
Explain the role of a DBMS	Describe the software that controls a database.
Compare storage options	Choose between HDD, SSD, USB, memory card, NAS, and cloud storage.
Use productivity tools appropriately	Choose a word processor, spreadsheet, presentation tool, form, calendar, or collaboration app for a task.
Connect the three pillars	Trace what happens from data entry to saving, storage, backup, and sharing.

KEY POINT The easiest way to remember the whole lesson

Database = organizes the information. Storage = keeps the information. Productivity tool = helps a person create, view, analyze, or share the information.

The Three Pillars of Digital Work

Almost every task you do on a computer involves these three layers, even when you do not notice them.

1. You enter or create data	2. A database organizes it	3. Storage keeps it	4. A tool shows or uses it	5. People act on it
-----------------------------	----------------------------	---------------------	----------------------------	---------------------

REAL-WORLD EXAMPLE School enrollment

A student fills out an online enrollment form. The form is a productivity tool. The submitted name, program, year level, and contact details are organized in a database. The database files themselves are stored on a server or cloud storage. Later, staff use another productivity tool to search, print, or report the enrollment data.

REAL-WORLD EXAMPLE Ordering food through an app

You choose a meal, quantity, address, and payment method. The app saves your order details in a database. The server stores the database. The restaurant dashboard reads that data and shows the staff what to prepare.

TRY THIS One-minute check

When you save a class activity in Google Docs, identify: (1) the productivity tool, (2) where the file is stored, and (3) what kind of organized information may be kept behind the scenes, such as the file owner, title, date modified, and sharing permissions.

What Is a Database?

A database is an organized collection of information designed so data can be stored, found, changed, and used efficiently.

SIMPLE IDEA Think of a database as a very organized digital filing cabinet

A normal folder can hold files, but a database stores information in a structured way. Because the structure is consistent, software can search thousands or millions of entries much faster than a person opening files one by one.

REAL-WORLD EXAMPLE Registrar's student list

Imagine 5,000 student records. If each student had a separate random text file, finding one student would be slow. In a database, the system can search a Student ID and return the matching record in a fraction of a second.

StudentID	Name	Program	YearLevel	Email
2026-001	Ana Cruz	BSIT	1	ana@example.edu
2026-002	Mark Lee	BSBA	2	mark@example.edu
2026-003	Rina Santos	BSIT	1	rina@example.edu

The table above is only a tiny example. A real school database may contain many related tables for students, subjects, grades, payments, instructors, schedules, and more.

Tables, Records, and Fields

These three words appear constantly in database discussions. They are easier than they sound.

Term	Simple meaning	Example in the student table
Table	A collection of related data arranged in rows and columns.	The entire list of students.
Record	One complete entry, usually one row.	The row for Ana Cruz.
Field	One type of information, usually one column.	Email, Program, or YearLevel.
Value	The actual content inside one cell.	BSIT or 2026-001.

SIMPLE IDEA A simple analogy

Think of a class list on paper. The whole class list is the table. One student's row is one record. The "Name" column is a field. "Ana Cruz" is a value inside that field.

TRY THIS Point it out

Using the sample table on the previous page: Which row is one record? Which column is one field? What is the value stored in Rina Santos's Program field?

Why Databases Use Keys

Relational databases connect tables using shared identifiers called keys. This prevents the same information from being copied everywhere.

Students table		Enrollments table
StudentID	Name	StudentID SubjectCode
2026-001	Ana Cruz	2026-001 IT101
2026-002	Mark Lee	2026-002 BUS102
2026-003	Rina Santos	2026-003 IT101

SIMPLE IDEA Why use StudentID instead of only a name?

Names can be repeated. Two students might both be named “John Reyes.” A unique StudentID gives the system a safer way to identify the correct person.

KEY POINT Primary key

A primary key is a field whose value uniquely identifies a record, such as StudentID, EmployeeID, OrderID, or ProductID.

REAL-WORLD EXAMPLE Online shopping

An Orders table can store OrderID and CustomerID. The CustomerID points to the correct customer record. The system does not need to copy the customer’s full profile into every order row.

Relational (SQL) vs. NoSQL

Databases can be organized in different ways. Two widely used approaches are relational databases and NoSQL databases.

Relational (SQL)	NoSQL
Usually uses tables with defined columns.	May use documents, key-value pairs, graphs, or other flexible structures.
Best when the data format is predictable.	Useful when the data shape changes often or varies widely.
Strong fit for transactions and consistent records.	Often chosen for flexible, large-scale, or rapidly changing applications.
Examples: MySQL, PostgreSQL, Oracle.	Examples: MongoDB, Redis, Cassandra.

REAL-WORLD EXAMPLE When SQL makes sense

A school's official grades usually follow a predictable structure: StudentID, SubjectCode, Term, Grade, SchoolYear. A structured relational table is a natural fit because consistency matters.

REAL-WORLD EXAMPLE When flexible data can help

A content platform may store posts that have different combinations of text, images, tags, reactions, and metadata. A document-style NoSQL database can handle changing structures more flexibly.

COMMON MISTAKE NoSQL does not mean “no structure”

NoSQL systems still organize data. They simply do not rely only on the traditional fixed relational-table model.

What Is a DBMS?

A Database Management System (DBMS) is the software layer that creates, controls, secures, and manages a database.

DBMS job	What it means in real life
Controls access	A student may view only their own grades, while an instructor may enter grades for assigned classes.
Keeps data consistent	If two users update information at the same time, the DBMS helps apply changes correctly.
Protects against loss	Backups and recovery mechanisms help restore data after errors or failures.
Optimizes performance	Indexes and query planning help the system find records faster.
Enforces rules	The system can reject invalid or duplicate values when rules require them.

SIMPLE IDEA Database vs. DBMS

The database is the organized data. The DBMS is the software that manages that data. Example: your student records are data; MySQL can be the DBMS managing those records.

Common DBMS examples include MySQL, PostgreSQL, Microsoft SQL Server, Oracle, MongoDB, and SQLite.

How Software Finds Data

Users usually do not search raw database files manually. Applications send requests called queries to the DBMS.

User asks for data	Application creates a query	DBMS searches the database	Matching records return	Application displays results
--------------------	-----------------------------	----------------------------	-------------------------	------------------------------

REAL-WORLD EXAMPLE Checking a grade

A student opens “My Grades.” The application may request records where StudentID matches the logged-in student and SchoolYear matches the selected term. The DBMS returns only the matching grade records, and the page displays them.

SIMPLE IDEA You do not need to memorize SQL yet

At this level, focus on the idea: a query is a structured request for data. It can ask to find, add, update, summarize, or remove records depending on permissions.

TRY THIS Think like a system

If a librarian wants to see all books that are currently borrowed, what information should the database search for? Example answer: records whose status is “Borrowed” or whose return date is still empty.

Where Data Actually Lives

A database organizes information, but the information still needs physical or remote storage. Storage systems keep files and data available now and later.

SIMPLE IDEA Database and storage are not the same thing

A database is an organizational system for data. Storage is the place or medium where the database files and other files are actually kept.

REAL-WORLD EXAMPLE Your phone

The Photos app organizes pictures so you can search and view them. The phone's internal flash storage physically holds the image files. If cloud backup is enabled, copies may also exist on remote servers.

Storage category	Examples	Main idea
Local storage	HDD, SSD, USB flash drive, memory card	Physically attached to or inserted into a device.
Network storage	NAS or shared office drive	Stored on another device reachable through a local network.
Cloud storage	Google Drive, OneDrive, Dropbox, iCloud	Stored on remote servers and accessed through the internet.

HDD, SSD, USB Flash Drive, and Memory Card

All of these can store data, but they differ in speed, durability, size, cost, and typical use.

Device	How it works / feels	Good for	Limitations
HDD	Uses spinning magnetic disks and moving parts.	Large storage at lower cost.	Slower and more sensitive to physical shock.
SSD	Uses flash memory with no moving parts.	Fast startup, applications, and daily computer use.	Usually costs more per gigabyte than HDD.
USB flash drive	Small portable flash storage.	Moving files between devices.	Easy to lose; not ideal as the only backup.
Memory card	Tiny removable flash storage.	Phones, cameras, action cameras, some laptops.	Small size makes it easy to damage or misplace.

REAL-WORLD EXAMPLE Choosing for a school computer lab

An SSD is a strong choice for the operating system and applications because it makes computers feel faster. A larger HDD may still be useful for inexpensive bulk storage, depending on the setup.

COMMON MISTAKE Portable does not mean safe

A USB flash drive is convenient, but it can fail, be lost, or be infected by malware. Important files should have another backup copy.

STORAGE CAPACITY

KB, MB, GB, and TB

Storage capacity tells you approximately how much data a device can hold. The numbers become easier when you connect them to familiar files.

Unit	Very simple idea	Typical example
KB - Kilobyte	Small amount of data	A small text file may use only a few KB.
MB - Megabyte	Larger than KB	A photo or short document may use several MB.
GB - Gigabyte	Much larger	Phones and laptops are commonly sold with storage measured in GB.
TB - Terabyte	Very large	External drives and servers often use TB-scale storage.

SIMPLE IDEA Do not obsess over exact file sizes

Actual sizes vary by quality, format, compression, and content. The important idea is the scale: KB < MB < GB < TB.

TRY THIS Which is larger?

Arrange these from smallest to largest: 500 MB, 2 TB, 128 GB, 900 KB. Answer: 900 KB, 500 MB, 128 GB, 2 TB.

NAS, Cloud Storage, Sync, and Backup

Networked storage lets people access information beyond one device. However, “sync” and “backup” are not exactly the same thing.

Concept	Simple explanation	Example
NAS	A storage device shared over a local network.	An office department keeps shared files on one network storage box.
Cloud storage	Files live on remote servers and are accessed through the internet.	A student opens the same Google Drive file on a phone and laptop.
Sync	Keeps copies on devices/services updated with the latest changes.	Edit a document on a laptop, then see the new version on a phone.
Backup	Keeps extra copies so data can be recovered after loss.	A server creates a daily copy of its database.

COMMON MISTAKE Sync is not always a full backup

If a synced file is accidentally deleted, that deletion may also sync. A proper backup strategy keeps recoverable copies or version history so earlier data can be restored.

REAL-WORLD EXAMPLE Broken laptop

If your only copy of a research paper is on one laptop and the drive fails, the file may be lost. If another copy exists in cloud storage or an external backup, recovery is much easier.

A Simple Backup Mindset

The LMS lesson emphasizes regular backup, access control, version tracking, and consistent organization. A practical way to think about backup is to avoid depending on a single copy.

SIMPLE IDEA Simple backup rule

For important school or work files, keep the main working copy plus at least one separate backup. For more critical data, organizations often keep multiple copies in different locations or media.

REAL-WORLD EXAMPLE Student thesis files

Working copy on laptop + cloud copy + occasional external-drive copy is safer than keeping everything in one folder on one device.

Good habit	Why it matters
Automatic backups	People forget. Automation reduces the chance of having no recent copy.
Version history	You can recover an earlier version after a bad edit or accidental overwrite.
Clear folder names	You find the correct file faster and reduce duplicate “final-final-v3” files.
Access permissions	Only the right people can view or edit sensitive information.
Test recovery	A backup is useful only if it can actually be restored.

Software That Helps People Get Work Done

Productivity tools help people create, organize, communicate, plan, analyze, and share work more efficiently.

Category	What it helps you do	Examples
Word processing	Write letters, reports, essays, and documents.	Microsoft Word, Google Docs
Spreadsheets	Calculate, organize rows/columns, create charts, analyze data.	Microsoft Excel, Google Sheets
Presentations	Explain ideas visually using slides.	PowerPoint, Google Slides
Forms / data entry	Collect structured responses.	Microsoft Forms, Google Forms
Communication	Send messages, email, and hold online meetings.	Gmail, Outlook, Teams, Zoom, Slack
Task management	Track work, deadlines, and responsibilities.	Trello and similar tools
Scheduling	Manage dates, events, and shared availability.	Google Calendar, Outlook Calendar

KEY POINT The best tool depends on the task

Do not use a word processor for everything. Use the tool whose strengths match the job you need to do.

CHOOSING THE RIGHT TOOL

Which Tool Should I Use?

A good digital worker does not only know how to open software. They know which tool is appropriate for a specific problem.

Task	Best-fit tool type	Reason
Write a reflection paper	Word processor	Designed for paragraphs, headings, page layout, and printing.
Compute class averages	Spreadsheet	Formulas can calculate totals and averages automatically.
Present a project proposal	Presentation software	Combines short text, visuals, charts, and speaking points.
Collect 100 survey responses	Online form	Standardizes questions and stores responses automatically.
Assign tasks to group members	Task board / project tool	Shows who owns each task and its status.
Schedule a consultation	Calendar	Handles dates, times, reminders, and availability.

COMMON MISTAKE Using the wrong tool creates extra work

Typing 100 scores in a Word table and manually calculating averages is possible, but a spreadsheet is much better because formulas can do the calculation consistently.

Why Spreadsheets Are More Than Tables

A spreadsheet looks like rows and columns, but its main strength is calculation and analysis.

Student	Quiz 1	Quiz 2	Average
Ana	85	90	87.5
Mark	78	82	80.0
Rina	92	95	93.5

SIMPLE IDEA The important difference

A database table stores structured records for an application. A spreadsheet is a productivity tool people can use directly for calculations, analysis, and reporting. They can look similar, but their purpose and control are different.

REAL-WORLD EXAMPLE Class record

An instructor may use a spreadsheet to calculate grades, while the official school system stores final grades in a database. The spreadsheet helps with work; the database becomes the controlled system of record.

TRY THIS Tool choice

Would you use a database or a spreadsheet to manage official records for 20,000 students with logins, permissions, history, and multiple offices? A database-backed system is the better fit. For a quick personal calculation, a spreadsheet is usually simpler.

Working With Other People

Modern productivity tools let teams work on the same information without being in the same room.

Feature	What it solves
Shared documents	Several people can contribute to one file.
Comments	People can discuss a specific part without rewriting everything.
Version history	Changes can be reviewed and sometimes restored.
Messaging	Questions and updates move quickly.
Video meetings	Teams can discuss work remotely.
Shared calendars	Everyone can see schedules and important dates.
Task boards	Work can be assigned and tracked visibly.

REAL-WORLD EXAMPLE Group research project

The group writes in a shared document, stores datasets in a shared drive, schedules meetings in a calendar, communicates in a group channel, and tracks tasks on a board. These are all productivity and collaboration tools working around stored data.

COMMON MISTAKE Sharing everything with everyone is not collaboration

Good collaboration includes correct permissions. A teammate who only needs to read a final report may not need permission to delete or edit source files.

Scenario 1: Student Enrollment

Now combine the three pillars. Follow one student's information from entry to use.

Student completes form	System validates entries	Database stores student record	Server/cloud storage holds data	Staff dashboard displays record	Reports are generated
------------------------	--------------------------	--------------------------------	---------------------------------	---------------------------------	-----------------------

Step	Pillar involved	What is happening
1. Fill in form	Productivity / application tool	The student enters name, program, contact details, and other required data.
2. Validate	Application + DBMS	The system checks required fields and may reject invalid input.
3. Save	Database	A structured student record is created or updated.
4. Keep	Storage	The database files are physically stored on the server or cloud infrastructure.
5. View	Productivity / application tool	Registrar staff search, filter, or print student information.
6. Protect	Storage + DBMS controls	Backups, permissions, and recovery processes protect the information.

KEY POINT Notice the pattern

The productivity tool is what people interact with. The database organizes the records. Storage keeps the data available. All three are necessary for a complete information system.

Scenario 2: Small Shop Inventory

A shop owner wants to know what products are available, what was sold, and what needs restocking.

Information	Possible database field
Product identification	ProductID
Product name	ProductName
Current quantity	StockQty
Price	UnitPrice
Supplier	SupplierID
Last restock date	LastRestockDate

REAL-WORLD EXAMPLE A sale happens

The cashier scans ProductID P100. The system finds the product record, records the sale, and reduces StockQty. The database keeps the structured values. Storage holds the database. The cashier and manager see the result through productivity/application screens and reports.

SIMPLE IDEA Why accuracy matters

If the database says there are 10 items but the shelf has only 3, decisions become wrong. Good systems need correct data entry, clear procedures, and regular checking.

TRY THIS Design one field

What field would you add if the owner wants an automatic warning when stock becomes too low? One possible field is ReorderLevel. If StockQty is less than or equal to ReorderLevel, the system can flag the product.

Scenario 3: Online Class Submission

A student uploads an assignment through an LMS.

Student selects file	LMS records submission	File stored on server/cloud	Database stores metadata	Instructor opens submission	Feedback/grade saved
----------------------	------------------------	-----------------------------	--------------------------	-----------------------------	----------------------

What may be stored	Why it matters
StudentID	Identifies who submitted the work.
AssignmentID	Identifies which activity the file belongs to.
Filename or file path	Tells the system where the uploaded file is located.
Submission timestamp	Shows when it was submitted.
Status	Shows submitted, late, graded, returned, etc.
Grade / feedback	Stores assessment results.

SIMPLE IDEA Database data and file storage can be separate

The uploaded PDF itself may be stored as a file on disk or cloud storage, while the database stores information about that file, such as its owner, path, submission date, and activity.

Keeping Data Safe, Organized, and Useful

Good data practices reduce the risk of loss, confusion, unauthorized access, and accidental changes.

Practice	Good example	Bad example
Back up regularly	Automatic daily backup plus a separate recovery copy.	One copy on one laptop only.
Control access	Students view their own records; authorized staff update records.	Everyone shares one admin password.
Track versions	Use version history or dated revisions.	Overwrite the only copy repeatedly.
Organize consistently	Folders and filenames follow a clear naming rule.	Files named final2, final-new, final-real, final-use-this.
Use strong authentication	Unique passwords and multi-factor authentication when available.	Same weak password used everywhere.
Verify before deleting	Confirm the record/file and backup status first.	Delete important data without checking.

KEY POINT Security is not only an IT department job

Users protect data too. Careless sharing, weak passwords, accidental deletion, and poor file organization can cause real problems even when the technology itself is working correctly.

COMMON CONFUSIONS

Things Students Often Mix Up

These pairs sound similar, but they are not identical.

Do not confuse...	Difference
Database vs. spreadsheet	A database is designed for controlled, structured application data; a spreadsheet is a user-facing productivity tool for calculation and analysis.
Database vs. DBMS	Database = organized data. DBMS = software that manages the database.
Storage vs. database	Storage is where data physically lives; a database defines how data is structured and managed.
Cloud storage vs. internet	Cloud storage uses the internet to access remote servers; the internet itself is the network, not your storage space.
Sync vs. backup	Sync keeps copies updated; backup focuses on recoverability after loss or damage.
HDD vs. SSD	Both store files; SSD is generally faster and has no moving parts, while HDD often offers cheaper capacity.
SQL vs. NoSQL	SQL usually uses structured relational tables; NoSQL uses other flexible data models.

TRY THIS Say it in one sentence

Complete this: “When I submit a file in an LMS, the database _____, the storage system _____, and the productivity/application tool _____.”

The School Clinic Record System

Imagine a school clinic replacing paper logs with a digital system.

The nurse needs to record student visits, date and time, symptoms, action taken, and follow-up notes. The school wants the records organized, protected, searchable, and backed up.

A possible design

Part	Example
Database tables	Students, ClinicVisits, Users
Key fields	StudentID, VisitID, VisitDate, Complaint, ActionTaken
DBMS	A database management system controls records and permissions.
Storage	Server storage plus a backup location.
Productivity/application tool	Clinic form for data entry, search page, and printable report.
Access control	Nurse can enter records; authorized administrators may view summaries; students should not access other students' records.
Backup	Automatic scheduled backup with tested recovery procedure.

TRY THIS Your turn

If the clinic wants to know how many visits happened this month, which part helps most? The database stores the visit records; a query can count matching dates; a productivity/reporting tool displays the result.

Important Terms

Key terms used throughout the lesson are listed below for quick reference.

Term	Simple meaning
Database	An organized collection of information.
Table	Related data arranged in rows and columns.
Record	One complete entry or row.
Field	One category/type of data, usually a column.
Primary key	A field that uniquely identifies a record.
Query	A structured request to find or change data.
DBMS	Software that manages a database.
Relational database	A database that uses related tables and keys.
NoSQL	A group of non-relational database approaches with flexible data models.
HDD	Storage drive with spinning magnetic disks.
SSD	Fast storage using flash memory and no moving parts.
NAS	Shared storage device available through a local network.
Cloud storage	Storage on remote servers accessed through the internet.
Backup	An extra copy made for recovery.
Sync	Keeping copies updated across devices/services.
Productivity tool	Software used to create, organize, analyze, communicate, schedule, or share work.
Version history	A record of previous versions or changes.
Access control	Rules that decide who can view or change data.

SUMMARY

Key Ideas

Databases, storage systems, and productivity tools perform different roles, but they often work together in the same digital process.

SIMPLE IDEA 1. Databases organize information

They give data structure so it can be searched, updated, related, and trusted.

SIMPLE IDEA 2. Storage systems keep information

HDDs, SSDs, USB drives, memory cards, NAS, and cloud systems hold files and data so they remain available.

SIMPLE IDEA 3. Productivity tools put information to work

Documents, spreadsheets, presentations, forms, communication tools, calendars, and collaboration tools help people create, analyze, use, and share information.

KEY POINT The full idea

A digital system is useful because organized data is stored safely and then presented through tools that help people do real work.

Discussion: Choose one system you use every week - an LMS, social media app, online shop, banking app, or school portal. Identify how it uses a database, where its information is stored, and which tools allow people to use or manage that information.